Vectors I Homework 3: Vector Product

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Vector product: properties, parallel vectors, areas, collinearity via cross product. Q1 True/False from the Basics Worksheet (Vector Product). Q2–Q6 are full past-paper questions.

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\times\mathbf{a}=\mathbf{0}$.	
(b)	$\mathbf{a}\times\mathbf{a}=0$.	
(c)	$\mathbf{i}\times\mathbf{j}=\mathbf{k}$.	
(d)	$\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$.	
(e)	$\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$.	
(f)	$\mathbf{a}\times\mathbf{b}$ is a scalar.	
(g)	$\mathbf{k}\times\mathbf{j}=\mathbf{i}$.	
(h)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$.	
(i)	$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$.	
(j)	$(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$.	
(k)	$\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.	
(l)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar.	
(m)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.	
(n)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.	
(o)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.	
(p)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$.	
(q)	The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $\lvert\mathbf{a}\times\mathbf{b}\rvert$.	
(r)	The area of triangle OAB is $\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$.	
(s)	$\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector.	
(t)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$.	
(u)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$.	
(v)	$\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$.	
(w)	$\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.	
(x)	$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$.	

	Statement	T/F
(y)	The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $\lvert\overrightarrow{AB}\times\mathbf{d}\rvert$.	
(z)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.	

2 [ASRJC 2025 Prelim P2 Q1]

Relative to the origin O, the position vectors of two points A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively. The length of $\mathbf{a}$ is k units and $\mathbf{b}$ is a unit vector. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{6}$ radians. It is given that the point M lies on the line segment AB such that $AM:AB$ is $3:4$. Find the exact area of ΔOAM, giving your answer in terms of k. [4]

3 [EJC 2025 Promo Q3]

(a) Given that $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, what can be deduced about the vectors $\mathbf{a}$ and $\mathbf{b}$? [2]

(b) Find a unit vector $\mathbf{n}$ such that $\mathbf{n}\times(2\mathbf{i}-\mathbf{j}+3\mathbf{k})=\mathbf{0}$. [1]

(c) Find the cosine of the acute angle between $2\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and the y-axis. [2]

4 [NYJC 2025 Prelim P1 Q3]

(a) A triangle ABC is such that $AB^2+BC^2=AC^2$. By considering $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$ and using the fact that $|\mathbf{v}|^2=\mathbf{v}\cdot\mathbf{v}$ for any vector $\mathbf{v}$, prove that $\angle ABC$ is a right angle. [4]

(b) In a triangle ABC, the point D divides AC in the ratio $\lambda:(1-\lambda)$, where $0<\lambda<1$. Let the position vectors of A, B, C and D be denoted by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ respectively. Show that the area of triangle ABD is given by $k|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}|$, where k is to be determined in terms of λ. [4]

5 [NJC 2025 Prelim P2 Q3]

With respect to an origin O, points P and Q have variable position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively, given by

$$\mathbf{p}=(\cos t)\mathbf{i}+(\sin t)\mathbf{j}-\mathbf{k}, \qquad \mathbf{q}=(\cos 2t)\mathbf{i}-(\sin 2t)\mathbf{j}+\alpha\mathbf{k},$$

where α is a real parameter and $0<t<\pi$.

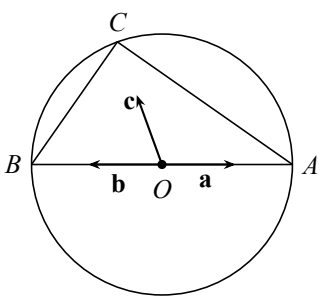
(i) By considering the cross product or otherwise, find the exact values of t and α if O, P and Q are collinear. [5]

It is given that $\alpha=0.5$.

(ii) Show that $\mathbf{p}\cdot\mathbf{q}=\cos 3t-0.5$. Find the greatest length of projection of $\mathbf{q}$ onto $\mathbf{p}$. [4]

6 [CJC 2025 Promo Q6]

The diagram shows a unit circle with centre O and diameter AB. The point C lies on the circumference of the circle. The position vectors of A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.



(a) By considering an appropriate scalar product, show that AC and BC are perpendicular. [4]

(b) Give a geometrical interpretation of $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$. [1]

The point D is such that $ABCD$ is a parallelogram. The point M is the mid-point of AD and the point R lies on OA produced such that $OA:AR=1:2$.

(c) By considering $\overrightarrow{CM}\times\overrightarrow{CR}$ or otherwise, show that the points C, M and R are collinear. [3]

END OF HOMEWORK